ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

ARTIFICIAL INTELLIGENCE

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Artificial Intelligence paper 1 · practice a.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (artificial intelligence). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct working even if a later step is wrong (consistent accuracy). Accept other correct examples and wording, and small rounding differences.

QUESTION 1

1.1 B ✓
1.2 B ✓
1.3 A ✓
1.4 B ✓
1.5 B ✓
1.6 B ✓
1.7 A ✓
1.8 B ✓
1.9 C ✓
1.10 B ✓

(10)

[10]

QUESTION 2

2.1 A rational agent chooses, for every sequence of percepts, the action that is expected to maximise its performance measure ✓, given what it has perceived and the knowledge it has built in ✓ (2)

2.2 Performance: safe, legal, fast, comfortable trips; profit; happy passengers ✓ (4)
Environment: roads and highways, other taxis and cars, pedestrians, traffic lights, potholes, weather, passengers ✓
Actuators: steering, accelerator, brakes, indicators, hooter, door control, display or speaker ✓
Sensors: cameras, lidar or radar, GPS, speedometer, odometer, engine sensors, microphone, payment card reader ✓

2.3 Partially observable, stochastic, sequential ✓ (3)
dynamic, continuous ✓
multi-agent ✓
(½ mark each, 6 dimensions)

2.4 (a) A simple reflex agent acts only on the current percept using condition–action rules; a model-based reflex agent also keeps an internal state (a model of the world) to handle what it cannot see now ✓✓ (3)
(b) A goal-based agent only asks whether an action reaches the goal; a utility-based agent compares how good different outcomes are (a utility value), e.g. the fastest and safest route ✓

[12]

QUESTION 3

3.1 Initial state: in town S. Actions: drive along a road to a neighbouring town ✓ (3)
Goal test: is the current town G? ✓
Path cost: the sum of the road distances driven ✓

3.2 Expansion order: S, A, B, C, D, E, G ✓✓ (4)
Path: S–A–C–G ✓ cost 3 + 4 + 9 = 16 ✓

3.3 Visit order: S, A, C, D, B, E, G ✓ (3)
Path: S–A–C–D–B–E–G ✓ cost 3 + 4 + 3 + 2 + 6 + 3 = 21 ✓

3.4 (5)

Expanded	Frontier after expansion (g)
S	A 3, B 5
A	B 5, C 7, D 9
B	C 7, D 7 (cheaper via B), E 11
C	D 7, E 11, G 16
D	E 11, G 13 (cheaper via D)
E	G 13
G	goal

Rows S and A ✓ row B with D updated to 7 ✓ rows C, D and E with G updated to 13 ✓
Path: S–B–D–G ✓ cost 5 + 2 + 6 = 13 ✓

3.5 Expansion order: S, B, E, G ✓ (3)
Path: S–B–E–G ✓ cost 5 + 6 + 3 = 14 (not optimal) ✓

3.6 (5)

Expanded	Frontier after expansion (f = g + h)
S	A 3 + 9 = 12, B 5 + 7 = 12
A	B 12, D 9 + 5 = 14, C 7 + 8 = 15
B	D 7 + 5 = 12 (updated), E 11 + 3 = 14, C 15
D	G 13 + 0 = 13, E 14, C 15
G	goal

Rows S and A ✓ row B with D updated ✓ row D ✓
Order S, A, B, D, G; path S–B–D–G ✓ cost 13 ✓

3.7 Yes: the true shortest distances to G are S 13, A 12, B 8, C 9, D 6, E 3, so h(n) never overestimates ✓ (2)
With an admissible heuristic (and a consistent one for graph search), A* is guaranteed to find the optimal (cheapest) path ✓

[25]

QUESTION 4

4.1 B = min(4, 9, 6) = 4 ✓ C = min(3, 7, 10) = 3 ✓ D = min(8, 11, 5) = 5 ✓ (4)
A = max(4, 3, 5) = 5, so MAX moves to D ✓

4.2 Not evaluated: 7 ✓ and 10 ✓ (the second and third leaves of C) (5)
After B, MAX is sure of at least 4 (α = 4) ✓
The first leaf of C is 3, so MIN can force C ≤ 3 ✓
C can never be better for MAX than B, so the rest of C cannot change the decision and is cut; D's leaves must all be checked because 8 and 11 are above 4 ✓

4.3 The game tree is far too big to search to the end of the game in the time allowed ✓, so the search stops at a depth limit and an evaluation function estimates how good each position is (e.g. material and piece positions) ✓ (2)

4.4 Expectiminimax adds chance nodes whose value is the probability-weighted average (expected value) of their children ✓; it is used for games with dice or cards, such as backgammon or Monopoly ✓ (2)

[13]

QUESTION 5

5.1 Clauses: ¬P ∨ Q, ¬Q ∨ R, P ✓ (5)
Add the negated goal ¬R ✓
Resolve ¬R with ¬Q ∨ R to get ¬Q ✓
Resolve ¬Q with ¬P ∨ Q to get ¬P ✓
Resolve ¬P with P to get the empty clause: a contradiction, so the KB entails R ✓

5.2.1 ∀x (Student(x) ∧ Studies(x) ⇒ Passes(x)) ✓ (1)

5.2.2 ∃y (Lecturer(y) ∧ ∀x (Student(x) ⇒ Likes(y, x))) ✓ (1)

5.2.3 ∀x (Robot(x) ⇒ ¬Student(x)) or ¬∃x (Robot(x) ∧ Student(x)) ✓ (1)

5.3 P(Ph | F) = P(F | Ph) P(Ph) / P(F) ✓ (5)
P(F | Ph) P(Ph) = 0.95 × 0.02 = 0.019 ✓
P(F) = 0.95 × 0.02 + 0.04 × 0.98 ✓ = 0.019 + 0.0392 = 0.0582 ✓
P(Ph | F) = 0.019 / 0.0582 = 0.326 (about 33%) ✓

5.4 It assumes that all the features (e.g. the words in an email) are conditionally independent of each other, given the class ✓, which is seldom true in real data, but the classifier still works well in practice ✓ (2)

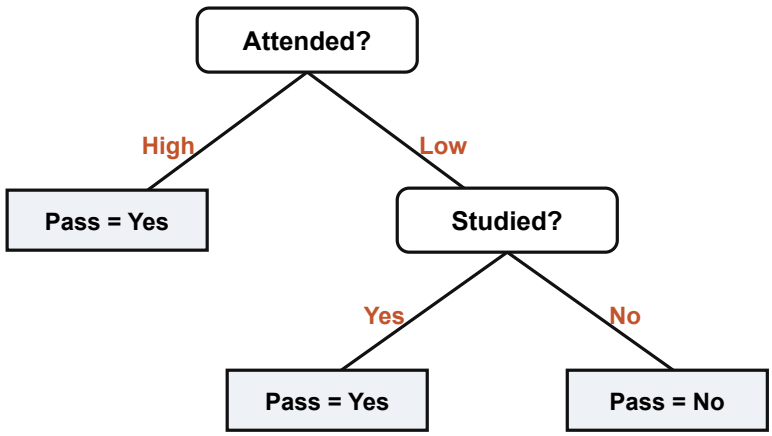
[15]

QUESTION 6

6.1.1 5 pass and 3 fail: $H(S) = -(0.625 \log_2 0.625 + 0.375 \log_2 0.375)$ ✓ = 0.424 + 0.531 = 0.954 ✓ (2)

6.1.2 Studied = Yes: 3 pass, 0 fail, H = 0 ✓ (6)
Studied = No: 2 pass, 3 fail, $H = -(0.4 \log_2 0.4 + 0.6 \log_2 0.6) = 0.971$ ✓
Gain(Studied) = 0.954 − (3/8 × 0 + 5/8 × 0.971) = 0.954 − 0.607 = 0.347 ✓
Attended = High: 4 pass, 0 fail, H = 0; Attended = Low: 1 pass, 3 fail, $H = -(0.25 \log_2 0.25 + 0.75 \log_2 0.75) = 0.811$ ✓
Gain(Attended) = 0.954 − (4/8 × 0 + 4/8 × 0.811) = 0.954 − 0.406 = 0.548 ✓
Attended has the higher gain, so it is the root ✓

6.1.3



Root Attended with High → Yes ✓ Low split on Studied ✓ (2)

6.2.1 P1 $\sqrt{13} = 3.606$ P2 $\sqrt{5} = 2.236$ P3 $\sqrt{10} = 3.162$ ✓ (2)
P4 $\sqrt{5} = 2.236$ P5 $\sqrt{18} = 4.243$ P6 $\sqrt{5} = 2.236$ ✓

6.2.2 k = 3: the nearest are P2 (A), P4 (B) and P6 (B) ✓, so X is classed as B ✓ (3)
k = 5: add P3 (A) and P1 (A): 3 A against 2 B, so X is classed as A ✓

6.3 $w_1x_1 + w_2x_2 + b = 0.2 - 0.4 + 0.1 = -0.1 < 0$, so the output is 0 ✓ (4)
Error = target − output = 1 − 0 = 1 ✓
$w_1 = 0.2 + 0.5 \times 1 \times 1 = 0.7$ and $w_2 = -0.4 + 0.5 \times 1 \times 1 = 0.1$ ✓
$b = 0.1 + 0.5 \times 1 = 0.6$ ✓

6.4 Accuracy = (40 + 130) / 200 = 0.85 ✓ (3)
Precision = 40 / (40 + 20) = 0.667 ✓
Recall = 40 / (40 + 10) = 0.8 ✓

6.5 Overfitting: the model learns the noise and details of the training data, so it does very well on the training data but badly on new data ✓ (3)
Any TWO ways: more training data ✓; a simpler model, e.g. pruning a decision tree or fewer features ✓; regularisation, early stopping, dropout, or cross-validation to choose the model

[25]

TOTAL: 100

END OF MARKING GUIDELINE